

Account Vending Machine design document

Blueprint Name: Account Vending Machine with MVC		
1	Blueprint overview	
1. 1.1	"As a customer, I can create new AWS accounts and assign it an Organizational Units (OUs) preconfigured with an account security baseline and a predefined network." (Scenario 1: Create a new AWS Account under MontyCloud Organization and deploy the MVC blueprint along with MC Onboarding Stack Scenario 2: On-Board a customer's Organization account and then deploy the Account Vending Machine setup into that Org Account and from there we can create New Accounts under the customer Org account and deploy the MVC along with DAY2 On-Boarding stack which will be an option.)	
1. 1.2	This Blueprint can quickly set up a secure, multi-account AWS environment based on AWS best practices and also help in saving time by automating the set-up of an environment for running secure and scalable workloads while implementing an initial security baseline through the creation of core accounts and resources. It also provides a baseline environment to get started with a multi-account architecture, identity and access management, governance, data security, network design, and logging. The AVM blueprint will create a new AWS Account and then deploy the MVC resources into that new account.	
1. 1.3	<i>This Blueprint solves the below pain points:</i> 1. Setting up a multi-account environment can take a significant amount of time, involve the configuration of multiple accounts and services, and require a deep understanding of AWS services. which is taken care of in this blueprint.	

2. Automatically setup the entire minimum viable infrastructure to have an efficient, secure and cost-effective AWS environment to work in.
3. It also helps in saving costs by restricting in over provisions of resources by enabling guard rails and billing alerts.
4. achieving Governance and Compliant environment by enabling monitoring, logging and config rules.

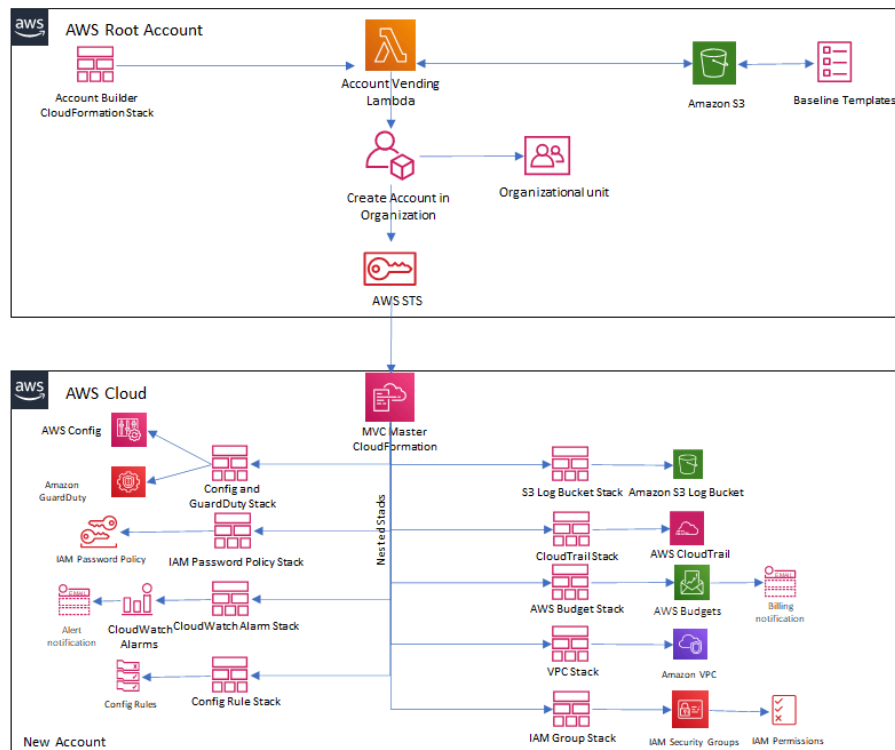
2 Blueprint category

2.1 Primary category: Infrastructure

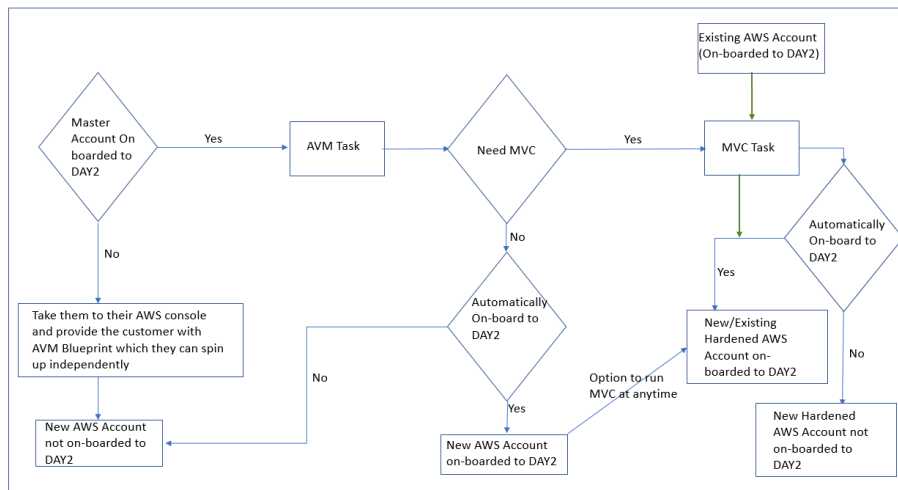
2.2 Secondary categories: MVC

3 Design diagram / flow diagram

Reference Architecture:



Flow Diagram



4 List of resources provisioned by the blueprint (Numbered list)

1. AWS Account
2. Organization SCP's
3. AWS Network baseline (VPC)
4. AWS Security baseline (Cloudtrail, Config, GuardDuty)
5. AWS IAM Security Roles(IAM Group, Password Policy)
6. Logging Components(CloudWatch Log, S3 to store Logs)
7. AWS Lambda Function (create New Account)
8. Alert Configurations (security alerts via SNS topic)
9. Billing Alerts (AWS Budget)

5 List of actions for each resource that the customers might take (Numbered list)

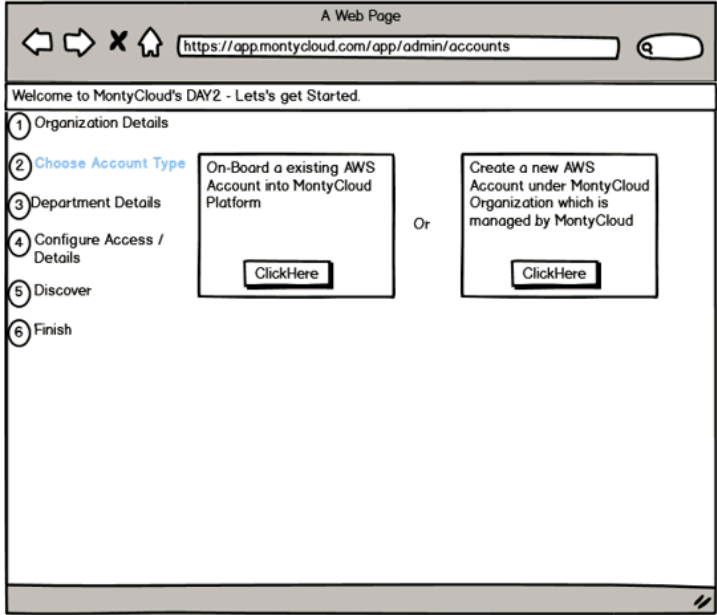
Resource: Action (Ex: EC2: Scan)

Enable VPC Peering/VPN Connection/Direct Connect (Shared Network)

6 List of monitored events/metrics that customers might want (Numbered list)

7 List of workflows/operations that we should consider for this blueprint

	1. As the First step deploys the CFN creates the Lambda and Custom resources to create a new AWS account and move it to a particular organization group and enable SCP policies. 2. And then in the second step deploy based on the options provided deploy the Minimum viable infrastructure within the new account and also provided a central logging option.	
8	Known limitations	
	1. New Accounts can be created only from the AWS Root Account.	
9	Source references (Be sure to list the source we are reusing or referring for our blueprints. It is critical that we give credits, mind licensing agreements, etc.)	
	1. https://aws.amazon.com/solutions/aws-landing-zone/ 2. https://medium.com/@superluminar/tested-for-you-multi-account-setups-with-aws-landing-zone-b934154cfc78	
10	Customization Options provided in Landing Zone and Control Tower	
	Control Tower When a new account is created using the AWS Control Tower account factory, the solution ensures that all resources attached to the account's OUs will be automatically deployed. Basically, the customization of CT helps in adding or editing the templates in the <code>custom-control-tower-configuration.zip</code> folder which is CT templates. But no access or visibility to these templates. Landing Zone The Customization of Landing Zone is defined by the templates configured in the Service Catalogue. We can add as many Service Catalog products based on the requirements a few mandatory products are deployed and few are deployed based on manual deployment in the Service Catalogue Product page.	
11	How are we Different from Control Tower	

	• Options to nuke all the Default Resources • Cost Control and billing alerts • Few Click deployment • No maintenance overhead in maintaining the Solution • Options to setup Account structure, which is not available in Control Tower, for example, Audit account, Security Account, and so on. • Granularity in setting up Security policies and SCP • Our Solution can work without SSO.	
12	Pain Points in Control Tower	
	• Very hard to maintain and configure • Need lots of AWS expertise for building the solution • Adding an existing account to CT is still a pain. • Offboarding from CT is very difficult. • CT is GA only in 4 regions	
13	Account Vending Machine with MontyCloud UI experience	
	Scenario 1: DAY 2 Created New AWS Account UI flow If the Customer request for a new AWS account under MontyCloud Organization below is a reference diagram for User experience to create a new account  AVM with disabled MVC Deployment	

A Web Page
https://app.montycloud.com/app/admin/accounts

Welcome to MontyCloud's DAY2 - Lets's get Started.

1 Organization Details
2 Choose Account Type
3 Department Details
4 **Configure Access / Details**
5 Discover
6 Finish

Configure New Account details

Account Email Address

Enter New Account Name

Admin Username

Admin user password

New Admin Role

☐ Enable Minimum Viable Cloud

Set up a secure public cloud environment to have a security baseline with maximum security protection, automated basic infrastructure deployment, log management and analytics, network traffic inspection and protection, cost control, alert management, and segmenting the network into multiple zones.

AVM with enabled MVC Deployment

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☒ Enable Minimum Viable Cloud

☐ Cleanup the default resources in the New AWS Account

☐ Create Custom IAM Roles and Groups for different departments in the New AWS Account

Set up a secure public cloud environment to have a security baseline with maximum security protection, automated basic infrastructure deployment, log management and analytics, network traffic inspection and protection, cost control, alert management, and segmenting the network into multiple zones.

Delete all the default resources in this new AWS account.

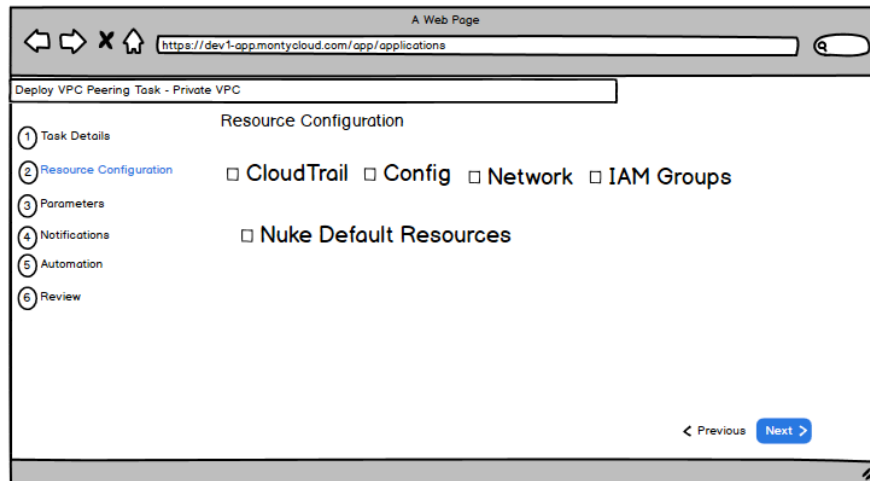
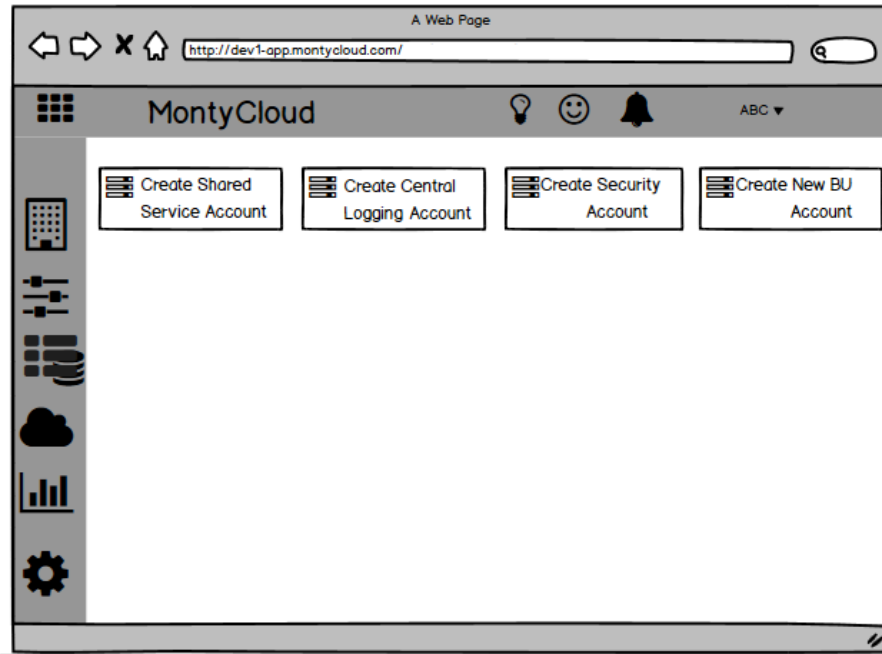
Required IAM permissions for Role and Groups, Cross-account access and setup standard IAM password policy

Network Information

VPC Name

Network Region

Network CIDR Range



14 Landing Zone Account Configuration

Security Account

- Security tools and audit
- Amazon GuardDuty
- AWS Security Hub
- AWS Config Aggregation / Compliance pack
- Cross-account roles

Log Archive Account

This account just holds the central S3 bucket to store logs for long-term retention. CloudTrail and Config services from all accounts are configured to store log files in this bucket by default. The bucket

should be protected by using the MFA-delete feature and should only be accessed for security audits or compliance investigations.	
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